

Opportunity Statement

We currently lack a consistent, event-based framework for measuring **user time-on-task** across workflows.

Without reliable telemetry, we cannot quantify how long users spend completing actions, where friction occurs, or how workflow design impacts productivity.

This PRD focuses solely on building the **technical infrastructure** to capture, transmit, and store standardized user interaction events that will later enable analytics teams to measure time-on-task accurately and consistently across applications.

Reference:  ROI Time on task instrumentation

Reference: [eRequest LifeCycle](#)

Goal

Create a scalable, event-driven logging infrastructure that captures discrete user interactions with the accuracy and completeness required to calculate time-on-task metrics.

Impact

- Provides a foundation for reliable time-on-task analytics across all major workflows.
- Enables BI and analytics teams to build dashboards and benchmarks on efficiency, throughput, and engagement.
- Reduces reliance on manual observation or guesswork.
- Establishes a standardized telemetry model for future analytics and optimization initiatives.

Hypotheses

- Implementing standardized event logging will enable >95% accuracy in time-on-task measurement compared to manual benchmarks.
- With event-driven timing data, we can identify at least three major workflow friction points within 30 days of live deployment.
- Creating a reusable telemetry pipeline will reduce instrumentation time for future analytics efforts by at least 50%.

Press Release (Future State)

For immediate release:

The Product Analytics team has launched a new **Event-Based Logging Infrastructure** to power precise measurement of user time-on-task across our ecosystem.

This new framework enables real-time insight into how users engage with workflows, allowing teams to identify friction, improve design, and quantify the impact of product changes.

This initiative marks a major milestone in moving from anecdotal understanding to measurable, data-driven workflow optimization.

Pre-Mortem (What Could Go Wrong and Why)

- **Over-logging:** Too many low-value events create data noise and unnecessary processing cost.
- **Schema inconsistency:** Different teams implement events differently, reducing usability.
- **Compliance issues:** Sensitive or user-identifiable data could be logged inadvertently.

Prioritized Scope

In Scope:

- Define and implement an event schema (event type, timestamp, session ID, user ID, metadata).
- Build an ingestion pipeline (e.g., Kafka, Snowflake, or equivalent) to collect and store events.
- Instrument key user actions (start, active, complete) in at least two representative workflows.
- Provide validation dashboards or logs to confirm event accuracy and data completeness.

Out of Scope:

- Downstream BI dashboards or time-on-task calculations.
- UX or interface changes.

- Historical backfill or data migration of previous logs.

Acceptance Criteria

- Events for at least two workflows are emitted, transmitted, and stored successfully.
- All events include a timestamp, user/session identifier, and standardized event type.
- Events appear in the analytics data store within 5 minutes of emission.
- Schema documentation and validation tests are complete and reviewed by analytics engineering.
- 100% compliance with data privacy and security standards.

Feature Type

Feature — foundational infrastructure supporting analytics enablement.

User-Facing?

No. This change is purely backend and analytics–infrastructure related.

Dependencies & Constraints

- **Dependencies:**
 - Platform Engineering for event transport and pipeline setup.
 - Analytics Engineering for schema, data model, and validation.
 - Security/Privacy teams for compliance review.
- **Constraints:**
 - Must align with existing logging frameworks.
 - Event data must not contain any personally identifiable information (PII).
 - Real-time ingestion latency must remain under 5 minutes.

Risks & Mitigation Strategies

- **Risk:** Event schema drift across products.
 - **Mitigation:** Establish a shared schema dictionary and validation rules.
- **Risk:** Event overload or excessive data volume.
 - **Mitigation:** Apply event sampling and limit to high-value user actions.
- **Risk:** Low adoption or inconsistent instrumentation.
 - **Mitigation:** Publish instrumentation SDK and documentation with clear examples.
- **Risk:** Data latency or ingestion failures.

- **Mitigation:** Set up Datadog alerts and SLA monitoring on event pipeline health.
- **Risk:** Compliance or data privacy violations.
 - **Mitigation:** Enforce PII redaction at ingestion layer and privacy reviews pre-launch.

Data & Analytics Plan

- **Inputs:** Discrete user interaction events (start, pause, resume, complete).
- **Outputs:** Structured event logs stored in analytics warehouse for time-on-task derivation.
- **Data Model:** `event_id`, `timestamp`, `user_id`, `session_id`, `event_type`, `metadata`.
- **Measurement:** Monitor event completeness and latency; validate event coverage versus manual logs.
- **Success Metrics:**
 - ≥95% event timestamp accuracy.
 - ≥90% coverage of target workflows.
 - <5-minute ingestion latency from emission to query readiness.

Open Questions

- How will user/session identifiers be unified across products?
- Should we include “idle” event tracking or focus only on explicit user actions?

Impact & Rollout Summary

Impact:

- Establishes the foundation for all time-on-task measurement and workflow performance analytics.
- Enables consistent behavioral data capture across products.
- Supports downstream efforts in efficiency benchmarking and ML-based productivity insights.

Rollout Strategy:

- **Phase 1 – Experiment:** Implement logging in two pilot workflows; validate data integrity and pipeline reliability.
- **Phase 2 – Alpha:** Expand to additional surfaces; finalize schema and compliance verification.

- **Phase 3 – General Availability (GA):** Standardize across all workflows; publish documentation and onboarding guides.